HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented structural analysis calculations for bridge components using **AutoCAD** and **Civil 3D**, adhering to project specifications and safety standards.
- Contributed to the development of sustainable engineering solutions by assisting in the investigation and optioneering phases for infrastructure projects, focusing on environmental impact assessments.
- Assisted in the preparation of technical documents, including **3D models** and design drawings, for civil structures, ensuring accuracy and compliance with client requirements.
- Supported bridge inspections and structural health monitoring activities by collecting and processing data from various sensors and assessment reports, contributing to asset management strategies.
- Developed carbon and cost estimates for infrastructure projects, working collaboratively with senior engineers to ensure realistic and achievable targets were set.
- Participated in Agile ceremonies and sprint planning sessions, providing input on technical estimations and task breakdowns for ongoing civil engineering projects.
- Containerized data processing scripts for structural analysis using **Docker**, facilitating consistent execution across development and testing environments on **AWS EC2**.

SKILLS

Civil Engineering – Structural Analysis, Design Calculations, Bridge Inspection, Asset Management, Concept Design, Detailed Design, Construction Supervision, Feasibility Studies, Optioneering, Carbon Estimates, Cost Estimates

Software & Tools – AutoCAD, Civil 3D, 3D Modelling, Microsoft Project, Primavera P6, Revit

Data & Analytics – Data Analysis, Structural Health Monitoring, Data Processing, Power BI, Tableau, DAX

Programming & Scripting – Python, R, SQL, Shell Scripting

Cloud & DevOps – AWS EC2, Docker, CI/CD, Git

Project Management & Methodologies – Agile, Scrum, Change Management, Technical Documentation

PROJECTS

Structural Assessment of Pedestrian Bridge

- Performed structural analysis using finite element methods in **AutoCAD** and **Civil 3D** to evaluate load-bearing capacity and stress points on a pedestrian bridge.
- Generated detailed technical drawings and 3D models for proposed reinforcement and repair strategies, contributing to the project's feasibility study.
- Documented the inspection findings and analysis results in a comprehensive report, highlighting areas requiring immediate attention and long-term maintenance planning.

Data Analytics for Infrastructure Asset Management

- Developed Python scripts to process and analyze sensor data from bridge structural health monitoring systems, identifying potential anomalies and performance degradation patterns.
- Created interactive **Power BI** dashboards to visualize key asset health indicators and historical performance trends for stakeholders, enabling data-informed maintenance decisions.
- Applied statistical methods to forecast potential future maintenance needs based on current asset condition and historical data, supporting proactive asset management.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)